Math 110 — Week 10

Solutions

HW10 P1. **Idea.** On $[-1,1]$, $\langle x^j, x^k\rangle = 0$ when $j+k$ is odd, so many of the Gram–Schmidt coefficients are 0.

1. $\|1\|^2 = \int_{-1}^{1} 1\,dx = 2$, so $e_1 = 1/\sqrt{2}$.
2. x is odd, so $\langle x, e_1\rangle = 0$ and $w_2 = x$. Then $\|x\|^2 = \frac{2}{3}$ and $e_2 = \sqrt{3/2}\,x$.
3. $\langle x^2, e_2\rangle = 0$ (odd integrand) and $\langle x^2, e_1\rangle = \frac{1}{\sqrt{2}} \cdot \frac{2}{3}$, so
$$w_3 = x^2 - \langle x^2, e_1\rangle e_1 = x^2 - \frac{2}{3}\cdot\frac{1}{2} = x^2 - \frac{1}{3}.$$
4. Its norm:
$$\|w_3\|^2 = \int_{-1}^{1}\Big(x^4 - \frac{2}{3}x^2 + \frac{1}{9}\Big)dx = \frac{2}{5} - \frac{4}{9} + \frac{2}{9} = \frac{8}{45}, \qquad e_3 = \sqrt{\frac{45}{8}}\Big(x^2 - \frac{1}{3}\Big) = \sqrt{\frac{5}{8}}\,(3x^2-1).$$
5. The orthonormal basis is $\frac{1}{\sqrt{2}}$, $\sqrt{\frac{3}{2}}\,x$, $\sqrt{\frac{5}{8}}(3x^2-1)$: multiples of the Legendre polynomials 1, x, $\frac{1}{2}(3x^2-1)$.

HW10 P2. **Idea.** $W^\perp$ is the null space of the matrix whose rows span W.

1. $x \in W^\perp$ if and only if x is orthogonal to both spanning vectors: $x_1 - x_2 + x_3 = 0$ and $2x_2 + x_3 = 0$. (x_4 is free.)
2. From the second equation $x_3 = -2x_2$; then $x_1 = x_2 - x_3 = 3x_2$. So $x = x_2(3,1,-2,0) + x_4(0,0,0,1)$.
3. Basis of $W^\perp$: $(3,1,-2,0)$, $(0,0,0,1)$. Check: $\dim W^\perp = 4-2 = 2$, and $(3,1,-2,0)$ is orthogonal to both spanning vectors ($3-1-2=0$, $2-2=0$).

HW10 P3. **Idea.** Write a general p with unknown coefficients and impose one equation; $W^\perp$ has dimension $3-1=2$.

1. For $p = a + bx + cx^2$, keep only the even terms of $p(x)(x+1)$, since odd terms integrate to 0 on $[-1,1]$:
$$\langle p, x+1\rangle = \int_{-1}^{1}\big(a + bx^2 + cx^2\big)dx = 2a + \frac{2}{3}b + \frac{2}{3}c.$$
2. So $p \in W^\perp$ if and only if $3a + b + c = 0$. Choosing $(a,b,c) = (1,-3,0)$ and $(0,1,-1)$ gives the basis $1-3x$, $x-x^2$ of $W^\perp$. They are independent (different degrees), and $\dim W^\perp = 2$.

HW10 P4. **Idea.** The easy rule is $(A+B)^\perp = A^\perp \cap B^\perp$. Apply it to $A = U^\perp$, $B = W^\perp$ and take complements, using $(X^\perp)^\perp = X$.

1. For any subspaces A, B: if x is orthogonal to every $a+b$, then taking $b=0$ shows $x \perp a$ for all $a \in A$, and taking $a=0$ shows $x \perp b$ for all $b \in B$. Conversely, if $x \perp A$ and $x \perp B$, then $\langle x, a+b\rangle = \langle x,a\rangle + \langle x,b\rangle = 0$. So $(A+B)^\perp = A^\perp \cap B^\perp$.
2. With $A = U^\perp$, $B = W^\perp$ and Prop. 90: $(U^\perp + W^\perp)^\perp = U \cap W$.
3. Take complements of both sides and use Prop. 90 again:
$$U^\perp + W^\perp = \big((U^\perp + W^\perp)^\perp\big)^\perp = (U \cap W)^\perp.$$

HW10 P5. **Idea.** Coefficients along an orthonormal basis are inner products; the closest point follows from Pythagoras.

(a) 1. Write $v = u + w$ with $u \in U$, $w \in U^\perp$ (Prop. 89), so $P_U v = u$.

2. Since $\langle w, e_j\rangle = 0$, $\langle v, e_j\rangle = \langle u, e_j\rangle$ for each j.

3. Prop. 86 in U gives
$$u = \sum_j \langle u, e_j\rangle e_j = \sum_j \langle v, e_j\rangle e_j.$$

(b) 1. For $u \in U$, $v - u = (v - P_U v) + (P_U v - u)$, with the first vector in $U^\perp$ and the second in U.

2. By Pythagoras,
$$\|v - u\|^2 = \|v - P_U v\|^2 + \|P_U v - u\|^2 \geq \|v - P_U v\|^2.$$

3. Equality holds if and only if $P_U v - u = 0$, that is, $u = P_U v$.

HW10 P6. **Idea.** Riesz representation: in an orthonormal basis, $p = \sum \psi(e_j) e_j$ (real scalars).

1. An orthonormal basis of $\mathcal{P}_1(\mathbb{R})$ for this inner product is $e_1 = 1/\sqrt{2}$, $e_2 = \sqrt{3/2}\, x$ (HW10 P1).

2. $\psi(f) = \int_0^2 f$ gives $\psi(e_1) = 2/\sqrt{2} = \sqrt{2}$ and
$$\psi(e_2) = \sqrt{3/2} \int_0^2 x\, dx = 2\sqrt{3/2}.$$

3. So
$$p = \sqrt{2} \cdot \frac{1}{\sqrt{2}} + 2\sqrt{\tfrac{3}{2}} \cdot \sqrt{\tfrac{3}{2}}\, x = 1 + 3x.$$

4. Check on the basis $1, x$:
$$\langle 1, 1 + 3x\rangle = \int_{-1}^1 (1 + 3x)\, dx = 2 = \int_0^2 1\, dx, \qquad \langle x, 1 + 3x\rangle = \int_{-1}^1 (x + 3x^2)\, dx = 2 = \int_0^2 x\, dx.$$

HW10 P7. **Idea.** ($\Rightarrow$) is Pythagoras. For ($\Leftarrow$), test the inequality on $v = u + tw$ with $u \in \operatorname{range} P$, $w \in \operatorname{null} P$: $Pv = u$ for every t, so $\|u\| \leq \|u + tw\|$ for all t, and a small t in the right direction would make $\|u + tw\|$ smaller than $\|u\|$ unless $\langle u, w\rangle = 0$.

1. ($\Rightarrow$) If $P = P_U$, then
$$\|v\|^2 = \|P_U v\|^2 + \|v - P_U v\|^2 \geq \|Pv\|^2.$$

2. ($\Leftarrow$) Let $u \in \operatorname{range} P$, $w \in \operatorname{null} P$ and $t \in \mathbb{F}$. Then $P(u + tw) = Pu + tPw = u$ (P fixes its range, Prop. 91), so the hypothesis gives
$$\|u\|^2 \leq \|u + tw\|^2 = \|u\|^2 + 2\operatorname{Re}\big(\bar{t}\langle u, w\rangle\big) + |t|^2\|w\|^2.$$

3. Take $t = -s\langle u, w\rangle$ with $s > 0$ real. Then $\bar{t}\langle u, w\rangle = -s|\langle u, w\rangle|^2$, and the inequality becomes $0 \leq -2s|\langle u, w\rangle|^2 + s^2|\langle u, w\rangle|^2\|w\|^2$. If $\langle u, w\rangle \neq 0$, divide by $s|\langle u, w\rangle|^2$ to get $2 \leq s\|w\|^2$ for every $s > 0$, which fails for small s. So $\langle u, w\rangle = 0$.

4. So $\operatorname{null} P \subseteq (\operatorname{range} P)^\perp$. By Prop. 91 and Prop. 89 both have dimension $n - \dim \operatorname{range} P$, so $\operatorname{null} P = (\operatorname{range} P)^\perp$.

5. With $U = \operatorname{range} P$: P fixes U and sends $U^\perp = \operatorname{null} P$ to 0, so $P(u + w) = u$, which is the definition of P_U.

O1. **Idea.** The minimizer is $P_U(x^2)$ for $U = \mathcal{P}_1$ in the L^2 product on $[0,1]$; use the orthonormal basis $1, \sqrt{3}(2x-1)$ of Ex. 70.

1. $\langle x^2, 1\rangle = \frac{1}{3}$ and $\langle x^2, \sqrt{3}(2x-1)\rangle = \frac{\sqrt{3}}{6}$ (computed in Ex. 70).
2. So
$$p = \frac{1}{3} + \frac{\sqrt{3}}{6} \cdot \sqrt{3}(2x-1) = \frac{1}{3} + x - \frac{1}{2} = x - \frac{1}{6}.$$
3. The minimum is $\|x^2 - p\|^2 = \|x^2\|^2 - \|p\|^2$ (Pythagoras), with $\|x^2\|^2 = \frac{1}{5}$ and $\|p\|^2 = \frac{1}{9} + \frac{1}{12} = \frac{7}{36}$, so the minimum is $\frac{1}{5} - \frac{7}{36} = \frac{1}{180}$. (This is $\|w_3\|^2$ in Ex. 70, because $x^2 - p = w_3$.)

O2. **Idea.** U is a line, so use the line formula; $U^\perp$ is the null space of the single row $(1,1,1,1)$; the distance is the length of the remainder.

1. Project onto the line (formula for $P_{\mathrm{span}(x)}$):
$$P_U v = \frac{\langle v, (1,1,1,1)\rangle}{\|(1,1,1,1)\|^2}(1,1,1,1) = \frac{10}{4}(1,1,1,1) = \left(\tfrac{5}{2}, \tfrac{5}{2}, \tfrac{5}{2}, \tfrac{5}{2}\right).$$
2. $U^\perp = \{x : x_1 + x_2 + x_3 + x_4 = 0\}$, with basis $(-1,1,0,0)$, $(-1,0,1,0)$, $(-1,0,0,1)$.
3. The distance is
$$\|v - P_U v\| = \left\|\left(-\tfrac{3}{2}, -\tfrac{1}{2}, \tfrac{1}{2}, \tfrac{3}{2}\right)\right\| = \sqrt{\tfrac{9}{4} + \tfrac{1}{4} + \tfrac{1}{4} + \tfrac{9}{4}} = \sqrt{5}.$$

O3. **Idea.** Write $v = u + w$ with $u \in U$, $w \in U^\perp$; then R flips the sign of u and keeps w.

1. $Rv = u + w - 2u = -u + w$. By Pythagoras, $\|Rv\|^2 = \|u\|^2 + \|w\|^2 = \|v\|^2$.
2. $R(Rv) = R(-u + w) = u + w = v$, so $R^2 = I$.
3. If U is a line in $\mathbb{R}^2$, R reverses the component along U and keeps the component along the perpendicular line $U^\perp$: it is the reflection across $U^\perp$.

O4. **Idea.** $\sum |\langle v, e_j\rangle|^2$ is $\|P_U v\|^2$ for $U = \mathrm{span}(e_1, \ldots, e_m)$, and the gap in Bessel's inequality is $\|v - P_U v\|^2$.

1. By Prop. 92 and Prop. 84, $P_U v = \sum_j \langle v, e_j\rangle e_j$ and $\|P_U v\|^2 = \sum_j |\langle v, e_j\rangle|^2$.
2. By Pythagoras, $\|v\|^2 = \|P_U v\|^2 + \|v - P_U v\|^2$, so
$$\|v\|^2 - \sum_j |\langle v, e_j\rangle|^2 = \|v\|^2 - \|P_U v\|^2 = \|v - P_U v\|^2.$$
3. So equality holds in Bessel's inequality if and only if $v - P_U v = 0$, that is, $v = P_U v$. If $v = P_U v$ then $v \in U$. Conversely, if $v \in U$, then $v = v + 0$ with $0 \in U^\perp$, so $P_U v = v$.

O5. **Idea.** The best $a + bt$ is the projection of $y = (1,2,4)$ onto $\mathrm{span}\big((1,1,1),(0,1,2)\big)$; the remainder must be orthogonal to both spanning vectors.

1. The remainder is $r = y - a(1,1,1) - b(0,1,2)$. The conditions $\langle r, (1,1,1)\rangle = 0$ and $\langle r, (0,1,2)\rangle = 0$ are
$$3a + 3b = 7, \qquad 3a + 5b = 10.$$
2. Subtracting, $2b = 3$, so $b = \frac{3}{2}$ and $a = \frac{5}{6}$. The line is $y = \frac{5}{6} + \frac{3}{2}t$.
3. Check: $r = \left(\frac{1}{6}, -\frac{1}{3}, \frac{1}{6}\right)$, which is orthogonal to $(1,1,1)$ and to $(0,1,2)$.

O6. **Idea.** Riesz representation with the orthonormal basis $e_1 = 1$, $e_2 = \sqrt{3}(2x-1)$, $e_3 = \sqrt{5}(6x^2-6x+1)$ of Ex. 70: $v = \sum e_j(0)e_j$.

1. $e_1(0) = 1$, $e_2(0) = -\sqrt{3}$, $e_3(0) = \sqrt{5}$.
2. So
$$v = 1 - \sqrt{3}\cdot\sqrt{3}(2x-1) + \sqrt{5}\cdot\sqrt{5}(6x^2-6x+1) = 1 - 3(2x-1) + 5(6x^2-6x+1) = 30x^2 - 36x + 9.$$
3. Check on $1, x, x^2$: $\int_0^1 v = 10 - 18 + 9 = 1$, $\int_0^1 xv = \frac{30}{4} - 12 + \frac{9}{2} = 0$, $\int_0^1 x^2 v = 6 - 9 + 3 = 0$, matching $p(0) = 1, 0, 0$.